hoBIT: A Profile-Aware Retrieval-Augmented Chatbot for University Academic Advising

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Code   Demo Video   Homepage

Abstract

In university academic advising, identical questions can require different answers depending on a student's department, admission cohort, and degree program, causing profile-blind retrievers to surface plausible but inapplicable evidence. We present **proFILL**, a method for transforming **hoBIT**—our college's current rule-based advising chatbot—into a profile-aware retrieval-augmented generation (RAG) system. Rather than requiring a complete user profile upfront, ProFILL progressively acquires only the profile attributes needed for each query, guided by both the query intent and the initially retrieved evidence, and uses them to condition retrieval over a profile-aware index. Extensive experiments and a human preference study show that ProFILL outperforms diverse RAG baselines, is preferred by target users, and remains effective with open-weight models for cost-effective on-premise deployment.

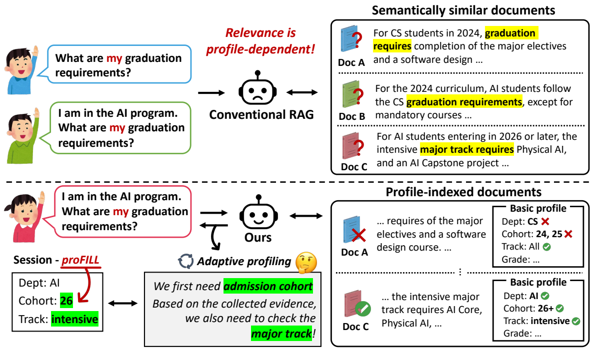

Figure 1: A conceptual comparison of (top) conventional RAG, which struggles with profile-dependent queries over semantically similar documents, and (bottom) the proposed hoBIT, which adaptively acquires the required profile information and retrieves the applicable document from a profile-indexed corpus.

1 Introduction

Retrieval-augmented generation (RAG) is increasingly employed in specialized domains (Gao et al., 2023b)—from medical education (Jang et al., 2025) to religious question answering (Gao et al., 2025). University academic advising is a practical and high-impact application. Students frequently ask questions about graduation requirements, required courses, double-major rules, and scholarship eligibility, yet an incorrect answer can directly affect course planning, delay graduation, or increase the administrative workload of advising offices.

The central difficulty is that academic advising questions are often *profile-dependent*. The correct answer is not determined by the query alone, but also by a student's profile, including department, admission cohort, major type, and related attributes. In our setting, students belong to one of three departments—Computer Science (CS), Data Science (DS), and Artificial Intelligence (AI)—where academic requirements and rules vary across admission years. Even within the same department, students may follow different rules depending on their major track, such as intensive, double-major, or convergence programs. Thus, the same question may require different reference documents for different students. Conventional RAG struggles in this setting, as semantically similar documents can lead a profile-blind retriever to return a plausible but wrong source (Figure 1). Hand-written FAQ rules are also hard to maintain, as profile-specific cases accumulate with policy revisions over time.

To address this gap, we present ProFILL, a method for transforming **hoBIT**—our college's current rule-based advising chatbot—into a profile-aware retrieval-augmented system. Our key idea is to treat profile-dependent evidence validity as part of the index structure, rather than as context to be resolved only at query time. We implement this idea through *offline profile-based indexing*, built on a lightweight academic profile schema covering department, admission cohort, major type, and re-

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lated attributes. During offline preprocessing, we use this schema to structure institutional materials, including regulations, department webpages, and notices. Each chunk is annotated with the profile values for which it is valid and the profile fields needed to interpret it. This turns the corpus into profile-conditioned evidence, allowing retrieval to distinguish not only what a document is about, but also which students it applies to.

At online query time, we use this indexed structure to acquire the student profile on demand. Instead of requiring login or a complete profile upfront, we maintain a time-limited session profile and collect only the fields needed by the current query. This is achieved via *on-demand adaptive-profiling*: we first infer the query intent, identify the necessary profile fields, and ask only for missing ones. The filled profile is combined with the query to match and filter against the indexed profile-conditioned evidence. We then inspect the retrieved evidence and ask a follow-up question only when fine-grained additional information is needed to verify applicability, such as detailed scholarship eligibility conditions. Once the user provides the additional field, the session profile is updated and retrieval is rerun with the completed profile. This evidence-driven selective process reduces initial user burden by avoiding a full upfront profile form, while improving answer correctness by triggering additional questions only when the retrieved evidence reveals unresolved profile requirements.

Contributions. **(1)** We present ProFILL, a method for transforming **hoBIT** into a profile-aware RAG system that replaces hand-written FAQ rules with profile-conditioned retrieval. **(2)** In ProFILL, we redesign both offline indexing and online query processing to address the profile-dependent nature of academic advising. **(3)** Extensive experiments show that ProFILL outperforms diverse RAG baselines, including profile-blind reranking and query-augmentation methods, while remaining effective across various open-weight models.

2 The hoBIT System

Overview. hoBIT is an academic-advising chatbot deployed at our college of informatics. Its original backend is a rule-driven dialogue system that maps user queries to predefined FAQ responses, limiting its ability to handle the profile-dependent questions common in academic advising. We extend hoBIT with ProFILL, a profile-aware retrieval-augmented generation framework featured with two key components: *offline profile-aware indexing* and *online on-demand profiling* (Figure 2).

The resulting system can support a range of advising needs, including curriculum and graduation requirements, notices, and career-related questions. The default profile schema comprises five attributes—`department`, `major type`, `grade`, `admission year`, and `student status`—that determine which curricula, policies, and other institutional information apply. The schema can be flexibly extended with additional attributes to support new advising needs.

2.1 Offline Profile-based Indexing

We build the corpus from five institutional sources: regulation and orientation PDFs, department webpages, board notices, and administrator-curated FAQs. After cleaning and chunking the collected materials, we use an LLM to annotate each chunk with a structured profile record over the five predefined attributes. Specifically, for each chunk, the LLM identifies which attributes determine the students to whom the information applies and fills in their corresponding values. Attributes that do not affect the chunk's applicability are left `null`. The resulting non-null fields encode the chunk's profile-dependent applicability and unveil which user attributes must be known for reliable retrieval.

In practice, we maintain separate static and dynamic indices based on source update frequency. Relatively stable materials (e.g., regulations) are stored in the static index, whereas frequently updated materials (e.g., notices) are stored in the dynamic index. Results from both indices are combined at query time. Further details on preprocessing and indexing are provided in Appendix A.

2.2 Online Query Processing

2.2.1 Intent Routing and Session-Profile Initialization

Intent Routing. At query time, an LLM first classifies each incoming query into one of five intents—*retrieval*, *ability*, *faq*, *greeting*, and *smalltalk*. Only *retrieval* queries proceed to subsequent retrieval and answer generation. *Greeting*, *ability*, and *faq* queries are handled using predefined templates or the FAQ store without an additional LLM generation call, while *smalltalk* queries receive a lightweight conversational response.

Session-Profile Initialization. For each retrieval

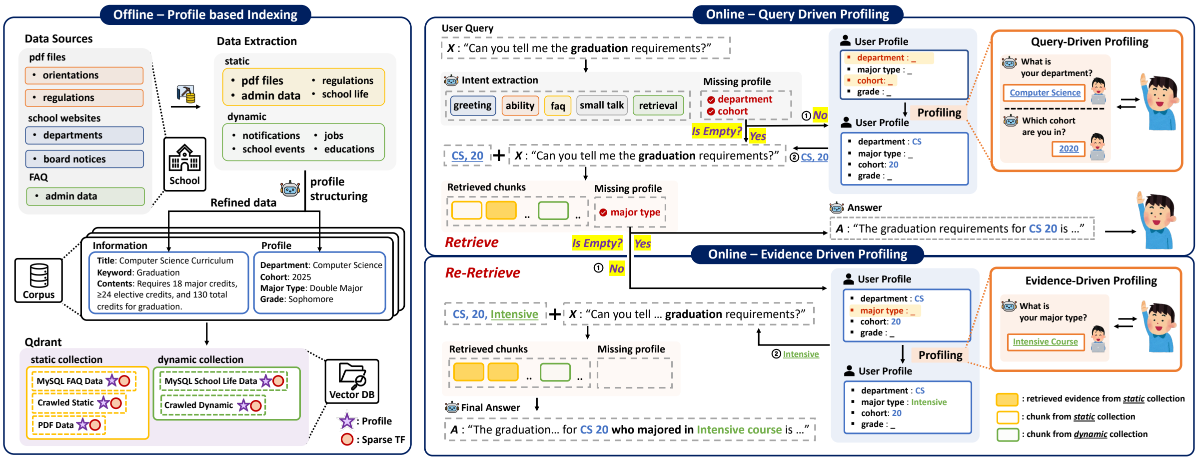

Figure 2: Overview of hoBIT. **Offline (left):** each chunk is annotated with its applicable profile values and required fields before indexing. **Online (right):** query-driven profiling acquires missing attributes for profile-aware retrieval, while evidence-driven profiling requests additional information and re-retrieves when needed.

query, the LLM identifies the profile attributes required to answer the question and extracts any corresponding values available from the query. These values initialize the session profile, while required attributes whose values are unavailable are marked as missing. These missing values are subsequently acquired as needed via adaptive profiling.

Importantly, the initialized profile serves only as a guide rather than a fixed specification of all information required to answer the query. Additional information that cannot be inferred from the query or is not covered by the predefined profile schema may be acquired on demand when its necessity becomes evident from the retrieved evidence.

2.2.2 On-demand Adaptive Profiling and Profile-Aware RAG

Rather than requiring login or a complete profile form upfront, ProFILL acquires only the information needed for the current query through two complementary stages: *query-driven profiling* and *evidence-driven profiling*.

Query-Driven Profiling and Retrieval. Before retrieval, the system asks the user for the attributes marked as missing during session-profile initialization. The acquired values are added to the session profile and incorporated into retrieval through both *soft* query augmentation and *hard* filtering. Specifically, the available profile values are serialized and prepended to the original query for retrieval.[1] The retrieved candidates are then filtered using the profile annotations: chunks whose non-null profile values conflict with the session profile are removed, while profile-compatible and profile-agnostic chunks remain eligible. The top ten eligible chunks are provided to the generator LLM.

Before generating an answer, an LLM selects the chunks needed to answer the query as the evidence set and checks whether their applicability can be determined from the current session profile. Specifically, it identifies any non-null profile fields of the selected chunks whose corresponding attributes remain unresolved, as these fields indicate the information required to determine whether each chunk applies to the user. The LLM also checks whether additional user information is needed to interpret the evidence reliably. If no further information is required, the system generates an answer from the selected evidence; otherwise, evidence-driven profiling is triggered.[2]

Evidence-Driven Profiling and Re-Retrieval. When additional information is required, the system asks the user a targeted follow-up question about the unresolved profile fields or other required information. The acquired information is added to the session profile, and retrieval is repeated in the same manner using the updated profile. An LLM then reselects the relevant evidence and generates the final answer from the refined evidence set.

2.3 System Implementation

To improve usability and user trust, hoBIT incorporates two interaction design features. First, for attributes covered by the predefined profile schema (e.g., `admission year`), the system presents predefined options instead of requiring free-form

[1]Further details on retrieval are provided in Appendix B.

[2]In our setup, this occurs for 24.3% of queries on average.

input, reducing user effort. Second, each generated answer is accompanied by the selected evidence, allowing users to directly inspect the institutional sources supporting the response.

3 Experimental Setup

Corpus. We construct the corpus from 515 institutional sources collected from our college: 3 academic-regulation and orientation PDFs, 89 department webpages, 137 board notices, and 286 administrator-maintained FAQ entries.

Profile-Grounded QA Data. With guidance from our college's academic affairs office and drawing on query logs from the deployed hoBIT service, we construct 1,800 QA instances spanning 60 unique student profiles, 10 advising categories,[3] and three query types: formal, first-person, and affirmative or negative verification queries. We additionally assess *intent-routing* accuracy and performance on *open-ended advising* questions without definitive ground-truth answers in Appendix D.

Evaluation Settings. We evaluate the profile-grounded QA under two settings. **Deployment** reflects the realistic scenario in which the user profile is initially unavailable and must be acquired during interaction. **Oracle** provides the complete profile along with the query, allowing us to assess how effectively ProFILL leverages the available profile information. For RAG baselines, the given profile is linearized and appended to the query as additional context. For all methods, we use `text-embedding-3-small` as the dense retriever and `gpt-4o-mini` as the generator. Latency is measured using a 48 GB MIG instance on a single NVIDIA RTX PRO 6000 GPU and an Intel Xeon Gold 6530 CPU.

Metrics. We evaluate retrieval using MRR and Recall@$\{1, 5, 10, 50\}$. Generation is evaluated using three metric types: (1) lexical metrics (ROUGE-L and Token-F1), which measure surface-level overlap with reference answers; (2) matching metrics (Keyword Match and Source Match), which assess whether the answer includes the required key information and uses the correct sources; and (3) the LLM-based metric (Grounded Correctness), which evaluates whether the answer is both correct and grounded in appropriate sources.[4]

4 Results and Discussion

Comparison with RAG Baselines. Table 1 compares ProFILL with RAG baselines using the same generator but different retrieval strategies. Hybrid combines BM25 and dense retrieval, HyDE (Gao et al., 2023a) augments the query with LLM-generated context, and Reranker applies a cross-encoder to refine the retrieved results.[5]

Overall, ProFILL achieves the strongest retrieval and generation performance. In the *deployment* setting, it outperforms all baselines in MRR, including those given complete profiles under the *oracle* setting (0.593 vs. 0.475). It also shows clear gains in lexical and matching metrics, including Source Match (0.749 vs. 0.412), and achieves the highest Grounded Correctness in both settings. Interestingly, adding a profile-blind reranker to ProFILL degrades performance, indicating that reranking without profile information can be harmful once profile applicability has already been incorporated into retrieval. ProFILL also remains efficient, requiring 6.2 seconds per query, compared with 6.9 seconds for HyDE. The 1.8-second gap from oracle ProFILL (4.4 seconds) reflects the additional cost of on-demand adaptive profiling.

Human Preference Evaluation. To evaluate user preference within hoBIT's actual target population, we conduct a blind pairwise study with 48 students from our college. Participants compare ProFILL against conventional dense retrieval-based RAG on ten profile-dependent questions (E). In Figure 3, ProFILL is preferred on all ten questions, achieving an aggregate non-tie win rate of 85.3% ($p < 0.001$). Preference is highest for graduation- and credit-related questions (93–98%; DS-01, AI-01, CS-02, CS-03), whose answers strongly depend on the student's profile, and lowest for the broader question on available major courses (58%; DS-02).

Ablation study. Table 2 reports ablations performance of ProFILL. Both profile-injection mechanisms are critical: removing the soft prefix or hard filter largely degrades the retrieval performance. Disabling re-retrieval with the evidence-driven profiling also consistently degrades performance. No-

[3] The categories cover major requirements, general education requirements, core general education, foundational courses, graduation requirements and projects, broad versus intensive majors, general education areas, department-specific major electives, graduation-credit breakdowns, and credit recognition for double majors.

[4] We use `deepeval` (Confident AI, 2024) and report scores averaged across `Qwen2.5-32B-Instruct` and `gpt-4o-mini`. Further details are provided in Appendix C.

[5] We use `BAAI/bge-reranker-v2-m3`.

		Retrieval					Generation					
	Retrieval System	MRR	R@1	R@5	R@10	R@50	ROUGE-L	Token-F1	Keyword Match	Source Match	Grounded Correctness	s/q
Deployment	BM25	0.089	0.017	0.118	0.267	0.728	0.139	0.167	0.564	0.412	0.412	4.0
	Dense	0.031	0.011	0.022	0.049	0.400	0.149	0.176	0.548	0.217	0.292	4.3
	Hybrid	0.061	0.009	0.069	0.152	0.698	0.146	0.173	0.574	0.380	0.395	4.3
	+reranker	0.111	0.027	0.165	0.322	0.698	0.152	0.177	0.621	0.391	0.425	4.6
	HyDE	0.061	0.013	0.057	0.141	0.689	0.147	0.175	0.586	0.367	0.403	6.9
	+reranker	0.103	0.025	0.156	0.300	0.690	0.151	0.174	0.613	0.379	0.422	7.2
	proFILL	**0.593**	**0.492**	**0.718**	**0.782**	**0.936**	**0.172**	**0.199**	**0.699**	**0.749**	**0.625**	6.2
	+reranker	0.420	0.271	0.561	0.691	**0.936**	0.168	0.193	0.665	0.694	0.596	6.4
Oracle	BM25	0.211	0.059	0.329	0.624	0.979	0.155	0.184	0.648	0.646	0.555	4.0
	Dense	0.475	0.306	0.693	0.871	0.994	0.171	0.200	0.725	0.723	0.617	4.3
	Hybrid	0.401	0.219	0.647	0.841	0.993	0.171	0.199	0.716	0.705	0.621	4.4
	+reranker	0.151	0.033	0.224	0.444	0.993	0.160	0.186	0.623	0.583	0.530	4.5
	HyDE	0.424	0.242	0.683	0.886	0.992	0.169	0.198	0.716	0.695	0.608	7.1
	+reranker	0.159	0.039	0.229	0.487	0.993	0.164	0.191	0.643	0.591	0.538	7.2
	proFILL	**0.780**	**0.674**	**0.929**	**0.980**	**1.000**	**0.179**	**0.207**	**0.733**	**0.847**	**0.676**	4.4
	+reranker	0.549	0.397	0.704	0.829	**1.000**	0.174	0.201	0.685	0.787	0.638	4.5

Table 1: Retrieval and generation performance. For all methods, `gpt-4o-mini` is used as the answer generator. *s/q* denotes end-to-end latency in seconds per query.

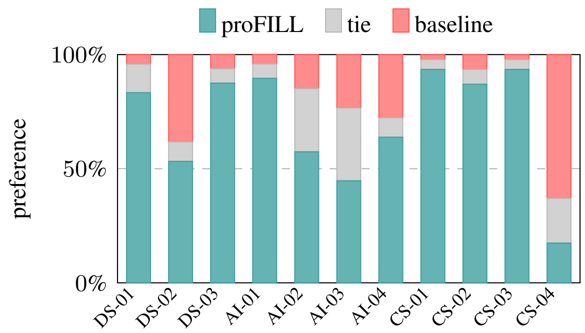

Figure 3: Human preference comparison. Results are reported as three-way vote splits. Question IDs are prefixed by the corresponding department.

Embedder	**Setting**	MRR	R@1	R@5	R@10	R@50
`openai_small`	proFILL	**0.596**	**0.495**	**0.721**	**0.784**	**0.936**
	w/o Hard filtering	0.311	0.157	0.502	0.671	0.923
	w/o Soft aug.	0.247	0.137	0.337	0.501	0.893
	w/o Re-retrieval	0.580	0.481	0.702	0.767	0.934
`BGE-M3`	proFILL	**0.665**	**0.579**	**0.775**	**0.796**	0.936
	w/o Hard filtering	0.475	0.349	0.624	0.704	0.934
	w/o Soft aug.	0.347	0.197	0.517	0.709	**0.938**
	w/o Re-retrieval	0.644	0.558	0.750	0.776	0.934
`Qwen3-Emb`	proFILL	**0.651**	**0.564**	**0.763**	**0.801**	**0.931**
	w/o Hard filtering	0.425	0.304	0.574	0.708	0.926
	w/o Soft aug.	0.304	0.184	0.408	0.556	0.919
	w/o Re-retrieval	0.636	0.549	0.748	0.785	0.927

Table 2: Ablation study. Results are reported with three different dense embedding models for retrieval.

tably, Recall@50 remains near ceiling without re-retrieval, indicating that its primary benefit is to promote relevant sources to higher ranks. This allows the generator to access the necessary evidence using fewer top-ranked chunks, thereby using its context window more efficiently. These trends are consistent across all three dense embedders, supporting the effectiveness of each design component.

Results with Varying LLMs. In Table 3, we evaluate ProFILL with various LLM generators

		Lexical		*Match*		
	Generator	ROUGE-L	Token-F1	Kw. Match	Src. Match	GC
Prop.	`gpt-4o-mini`	0.179	0.207	0.733	0.847	0.651
Open-weight	`Qwen3-8B`	0.169	0.197	0.721	0.879	0.670
	`Ministral-8B`	0.155	0.177	0.723	0.734	0.639
	`LLaMA3.1-8B`	**0.204**	**0.230**	0.665	0.817	0.649
	`EXAONE-7.8B`	0.135	0.158	0.704	**0.898**	**0.712**
	`Kanana1.5-8B`	0.123	0.143	**0.752**	0.861	0.697

Table 3: Results across various LLMs, including proprietary and open-weight models. GC denotes Grounded Correctness.

Open-weight models remain competitive with `gpt-4o-mini`, with `LLaMA3.1-8B` performing best on lexical metrics, `Kanana1.5-8B` on Keyword Match, and `EXAONE-7.8B` on Source Match and Grounded Correctness. These results demonstrate the viability of open-weight models for on-premise deployment.

5 System Demonstration

Figure 4 illustrates hoBIT's online interaction. We present two representative scenarios: a *cold start*, where the profile is acquired through *adaptive profiling*, and a *warm session*, where the cached profile is transparently reused.

UC1: Cold-start acquisition. A new student asks *"what are my required major courses?"* with no profile stored. hoBIT first acquires the student's `department` and `admission_year` through a *proactive* prompt, then requests `major_type` only when the retrieved curriculum requires it (*reactive*). The completed profile is cached, and the system retrieves the correct curriculum with sentence-level source citations.

UC2: Warm-session reuse. The same student next asks *"what are my graduation requirements?"*.

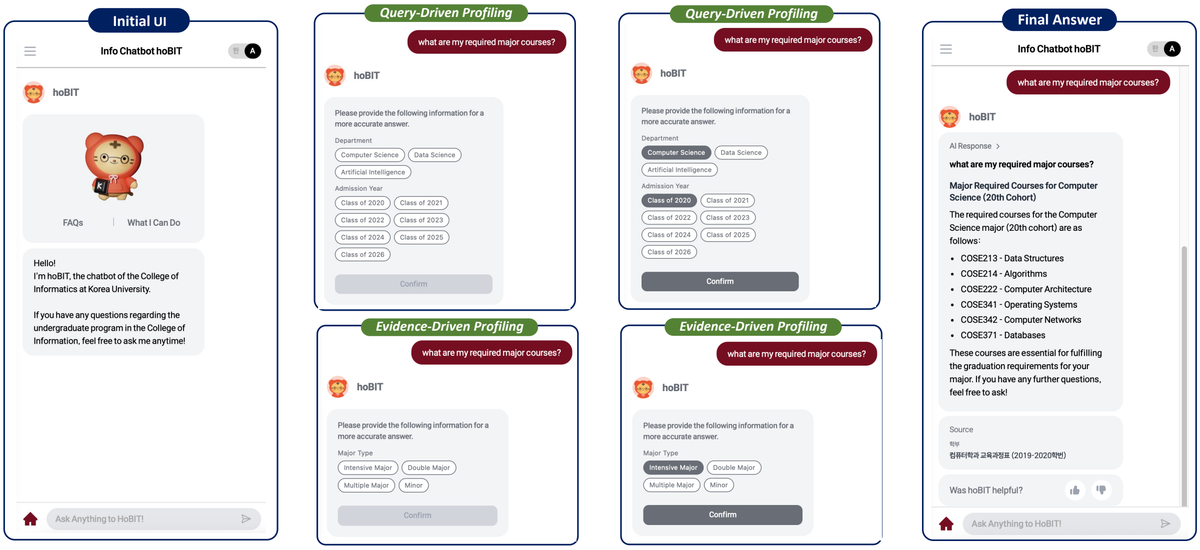

Figure 4: The hoBIT web interface: *adaptive profiling* acquires the profile at cold start (*proactive* then *reactive* ask-and-select), after which hoBIT answers with source citations and reuses the stored profile.

Because the profile is already cached, no further prompts are needed. The stored profile silently drives retrieval, returning the correct graduation requirements. The same query under a different profile retrieves a different curriculum.

6 Related Work

Retrieval-augmented generation. RAG (Lewis et al., 2020) grounds language-model generation in external evidence retrieved through dense (Karpukhin et al., 2020), lexical (Robertson and Zaragoza, 2009), or hybrid methods (Cormack et al., 2009). Its performance can be further improved through reranking (Nogueira and Cho, 2019), query augmentation (Gao et al., 2023a), structured indexing (Edge et al., 2024), or domain adaptation (Gao et al., 2023b; Jang et al., 2025). ProFILL focuses on the profile-dependent nature of academic advising by incorporating explicit user profiles into retrieval.

Personalization in LLMs. Various approaches have explored personalizing LLMs using user preferences (Choi et al., 2025; Thonet et al., 2025), interaction histories (Qin et al., 2025; Li et al., 2025), or personas (Zerhoudi and Granitzer, 2024). ProFILL instead targets personalization for academic advising through an explicit, schema-typed user profile. This structured profile is incorporated into retrieval through soft augmentation and hard filtering, without requiring per-user model training.

Academic-advising chatbots. Advising assistants have evolved from rule-based dialogue systems (Bocklisch et al., 2017) toward retrieval-grounded LLMs (Jang et al., 2025). The closest system, Marcel (Trienes et al., 2025), is not designed for profile-dependent advising and therefore returns the same answer to every student. In contrast, hoBIT uses adaptive profiling to provide answers tailored to each student's profile.

7 Conclusion

We present ProFILL, which transforms hoBIT from a rule-based chatbot into a profile-aware RAG system through offline profile indexing and on-demand adaptive profiling. It acquires missing information through query-driven profiling and requests additional details through evidence-driven profiling when needed. Experiments on institutional data provide practical insights into how structured profiles improve RAG for academic advising.

Limitations

(1) Single-institution scope. The corpus and benchmark come from one college of informatics; while proFILL's schema-typed profile and pipeline are domain-general, external validity across institutions and domains remains to be established. **(2) Constructed benchmark.** The profile-grounded track is built by crossing synthetic student cohorts with curriculum categories and phrasing styles, which isolates the profile signal under controlled conditions but does not capture the full variability of spontaneous student queries; the intent and open-ended tracks only partly mitigate this. **(3)**

Open-model latency. The open-weight generators are evaluated offline on a single MIG slice and do not yet serve live traffic, so reported latencies are indicative rather than production measurements. **(4) Self-reported profiles.** proFILL acquires profile fields from the student without institutional verification, so an incorrect self-report (e.g., a wrong admission year) propagates into retrieval; linking to authoritative records is future work.

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Appendix

Prompts are provided in our code repository.

A Offline Indexing

Each cleaned document is segmented into self-contained semantic chunks of up to 800 characters using paragraph-level splitting followed by sentence-level splitting. Before indexing, each chunk is prepended with its topic, subtopic, title, and extracted keywords to improve lexical matching. Expired notices are removed, and administrator-FAQ entries are consolidated. An LLM tagger (`gpt-4o-mini`) annotates each chunk over the five predefined profile attributes. For each attribute, it assigns the value or range of students to whom the chunk applies and sets unrelated attributes to `null`. The resulting non-null fields explicitly encode the chunk's profile-dependent applicability.

Chunks are divided by update frequency into a *static* collection for stable content and a *dynamic* collection for frequently updated notices. Each collection maintains sparse and dense indices for hybrid retrieval. The sparse index uses Kiwi-based Korean tokenization with content-morpheme filtering and 32K-dimensional feature hashing for BM25 scoring. The dense index uses `text-embedding-3-small` vectors.

B Retrieval Process

Given a query, we first translates non-Korean input into Korean. It then performs hybrid retrieval independently over the static and dynamic collections and combines their results through time-aware aggregation to construct the final retrieval results.

Hybrid Retrieval. Each collection is searched using both dense and BM25 retrieval, capturing overall semantic relevance and lexical term matching, respectively. The two rankings are combined using reciprocal rank fusion with $k{=}60$.

Time-Aware Aggregation. The system internally determines whether a query is time-sensitive and selects ten chunks accordingly. For time-insensitive queries, it selects seven results from the static collection and three from the dynamic collection; this allocation is reversed for time-sensitive queries. Within the dynamic collection, a recency score with a 90-day half-life favors newer notices among similarly relevant results.

C Evaluation Metrics and LLM Judges

Lexical and Matching Metrics. *ROUGE-L* and *Token-F1* measure overlap with reference answers after both texts are tokenized into Korean content morphemes using Kiwi, reducing penalties from inflectional variation. *Keyword Match* measures the proportion of expected content keywords included in the answer. *Source Match* evaluates attribution over the static and dynamic collections: a gold source cited in the answer scores 1, one retrieved but not cited scores 0.5, and a missing source scores 0, thereby rewarding grounded citation beyond retrieval alone.

LLM-judge criteria and prompts. Four judge metrics run through `deepeval`: *Contextual Precision* and *Contextual Recall* use deepeval's defaults over the retrieved context, while *Korean Faithfulness* and *Answer Correctness* are GEval metrics with domain-specific prompts. Generic judges misread Korean academic synonymy—a course code and its Korean name, or curriculum-era naming variants, denote the same entity—so every judge prompt is supplied a synonym-mapping table. The costlier judge metrics are scored on the 600-case subset stratified over the 10 categories and 3 phrasing types (60 per category); lexical and match metrics use all 1,800 cases. The two GEval prompts (rendered in English; Korean originals are used at run time) are:

> **Korean Faithfulness.** Score whether every factual claim in the answer is grounded in the retrieval context, treating domain synonyms as equivalent. An answer that declines ("information not found") with no factual claim scores 1.0 (vacuously faithful); an answer that fabricates facts when the context is empty or irrelevant scores 0.0. Output 0–1.

> **Answer Correctness.** *Step 1:* classify the question as positive-verification ("is X required?"), negative-verification ("X is not included, right?"), or general. *Steps 2–4:* score by type—for verification questions a correct yes/no on the target X scores 1.0 even as a one-word answer; for general questions, score by the fraction of expected key items (courses, credits, requirements) correctly present. *Step 5:* never penalize extra helpful information unless it is factually wrong. Output 0–1 (1.0 exact, 0.5 partial, 0.0 missing or incorrect).

D Dataset Construction and Supplementary Results

Profile-grounded QA. The index contains 906 chunks segmented and profile-annotated from the collected institutional sources (§2). The main

Intent class	Precision	Recall	F1	#Queries
greeting	1.000	0.630	0.773	100
ability	0.842	0.960	0.897	100
faq	0.971	1.000	0.985	100
smalltalk	0.704	1.000	0.826	100
retrieval	1.000	0.981	0.990	1,200

Table 4: Intent classification results.

dataset comprises 1,800 QA instances, covering all combinations of 60 student profiles, 10 profile-dependent advising categories, and three query types: formal, first-person, and verification-style. The 60 profiles combine 15 department–admission-year cohorts across CS, DS, and AI with four grade levels. Admission years are further grouped into eight curriculum-revision periods, which determine the applicable curriculum for each profile.

Each instance includes a deterministic gold source and expected keywords validated against the index. Profile information is provided only through the session profile and is excluded from the query text, isolating the effect of profile-aware retrieval. The advising categories were curated from the indexed materials and cross-checked against 797 query anchors distilled from 3,058 historical hoBIT logs, covering 14 of the 15 observed log categories.

Intent Routing. We construct 1,600 queries by combining 1,200 retrieval queries with 400 non-academic queries evenly distributed across *greeting*, *ability*, *faq*, and *smalltalk*. Only queries classified as *retrieval* enter the profile-aware RAG pipeline. As shown in Table 4, the classifier achieves an F1 score of 0.990 for retrieval queries and an overall accuracy of 0.960.

Open-ended Advising. We additionally evaluate open-ended advising questions covering 12 academic and student-life categories. Because these questions do not have deterministic reference answers, we evaluate Top-3 retrieval precision, Korean faithfulness (KO-F), and answer completeness using LLM judges (Table 5).

E Human Evaluation Details

Protocol. The blind pairwise study drew 48 participants—35 from Computer Science, 7 from Data Science, 2 from Artificial Intelligence, and 4 graduate or other (11 completed an English form, 37 a Korean form). Each of the 11 questions was shown as an A/B pair (proFILL vs. the Dense baseline) under the realistic *deployment* setting: proFILL acquires the needed profile fields on demand through adaptive profiling, while the Dense baseline is profile-blind. Presentation order was counterbalanced across participants (the proFILL/baseline order was reversed for DS-02 and CS-01), and each answer was labeled *proFILL*, *tie*, or *baseline*.

Category	Top-3	KO-F	Compl.	Avg	n
Academic status	0.713	0.942	0.973	0.876	100
Facilities/dining	0.683	0.898	0.931	0.837	100
Academic ops.	0.670	0.903	0.935	0.836	100
Scholarships	0.643	0.913	0.933	0.830	100
Space/facility	0.633	0.911	0.944	0.830	100
Enrollment/tuition	0.593	0.910	0.921	0.808	100
Clubs/council	0.610	0.881	0.909	0.800	100
Course reg.	0.593	0.878	0.926	0.799	100
Student services	0.567	0.894	0.935	0.799	100
Graduate/research	0.610	0.811	0.907	0.776	100
Notices/schedule	0.403	0.805	0.871	0.693	100
Career/internship	0.279	0.819	0.880	0.660	99
Overall	0.584	0.881	0.922	0.796	1,199

Table 5: **Open-ended advising by category** (Track III, 1,199; no ground truth). n: queries.

ID	Question	pF	tie	base	Win%
DS-01	Graduation requirements	40	6	2	95
DS-02	Available major courses	25	4	18	58
DS-03	Required major courses	42	3	3	93
AI-01	Graduation requirements	43	3	2	96
AI-02	GE-required courses	27	13	7	79
AI-03	Foundational courses	21	15	11	66
AI-04	Take required major courses?	30	4	13	70
CS-01	Recommend major courses	43	2	1	98
CS-02	Graduation requirements	40	3	3	93
CS-03	Credits to graduate	43	2	1	98
Total (10 discriminative)		354	55	61	**85.3**
CS-04†	List required major courses	8	9	29	22
Total (all 11)		362	64	90	80.1

Table 6: **Per-question human preference** (proFILL vs. Dense baseline). *pF*: proFILL votes; *Win%*: non-tie win rate. †CS-04 is non-discriminative—both systems returned the same correct course list—so it is excluded from the headline result.

F Deployment and Demonstration

GenAI Usage Disclosure

We use `gpt-4o-mini` as hoBIT's answer generator, as the LLM judge (via `deepeval`), and to generate the HyDE documents used in RQ2. Deterministic metrics are computed without LLM judgement. Prompt templates, evaluation settings, and the demonstration video are available on the project homepage. In addition, the authors used general-purpose AI assistants during paper drafting (organization and copy-editing) and as programming aids for portions of the implementation. All technical claims, system design decisions, experimental analyses, and final code review originate with and remain the responsibility of the authors.